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RESEARCH-ARTICLE

Nordic Perspectives on Algorithmic Systems: Cards as a Playful Intervention into the Crisis of Imagination

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Figure 1: The cover art of the Nordic Perspectives on Algorithmic Systems box (left), Caveats and Metaphors decks (center), and Settings and Methods decks (right). The box also contains a leaflet describing the contents and suggested uses.

Abstract

In this pictorial, we introduce a box with four card decks – focusing on settings, metaphors, methods, and caveats – designed to stimulate critical engagement with algorithmic systems from Nordic perspectives. We build upon the scholarship of Dumit [1, p. 604] who suggests that *"games are interesting tools because they involve the game player creatively within a dynamic system, requiring them to make decisions under constraints"*, and, therefore, capturing the systemic and dynamic nature of a socio-technical system, while positioning actors clearly into a particular structure. We apply this perspective to algorithmic systems as complex socio-technical assemblages that commonly entail emergent behaviors and dynamics. This leverages the fact that games excel at capturing dynamic, action-oriented, and inherently conflicted aspects of systems [2].

Today, algorithmic systems research often takes the form of critiquing systems-in-use. This leads to a *crisis of imagination*: rather than envisioning actively what algorithmic systems should be like, it is easy to feel hopeless and powerless amidst the problems of rapidly transforming digital societies. More broadly, dominant narratives and values – like the drive for scalability – dominate our attention. The effects of systems that have been scaled up globally, from search engines to social media and health records, are felt throughout societies. We use the notion of crisis of imagination to refer to the collective struggle to envision and articulate alternatives to seemingly inevitable AI-driven futures. *The Nordic Perspectives on Algorithmic Systems* card box is a tangible toolkit for responding to this crisis. It was created through workshops across Sweden, Denmark, and Finland, with the aim to destabilize hegemonic algorithmic narratives through situated and playful critique.

Reflecting on the observation that discussions about algorithmic systems quickly transform into discussions about society, we offer our card box as an artefact that can facilitate articulating positive and purposeful ideas about desirable societies and algorithmic systems which promote them. After detailed examples about each deck, we document two example games created by university students with inspiration from the card decks. *Just Sex* reimagines a digital contraception application as a morally uneasy board game where players navigate algorithmic advice amid societal gender biases. *YouTube Content Creation Game* exposes tensions between creator autonomy and platform opacity. Experiences indicate the cards can help questioning and reframing power dynamics and embedding situated values, while still remaining bounded by folk theories that dominate our understanding of algorithmic technologies.

Evidently, the Card Box does not 'solve' the crisis of imagination – not to even mention the polycrises surrounding us – but it serves to open room for shared reflection. Playful approaches are themselves a radical act at a time of crisis. By adopting Nordicness as a provocative caricature, we sidestep dominant technosolutionist framings to foster dialogues about what systems, values, and institutions we cherish and wish to sustain. The cards are a critical companion, a boundary object that researchers, designers, and other stakeholders can use to facilitate re-imagining algorithmic futures and alternatives.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**.

Keywords

algorithms, algorithmic systems, imagination, crisis, game design, Nordicness

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